

Network Analysis Report

Objective

To analyze network connectivity, inspect interface behavior, and simulate a network failure by disabling and enabling network interfaces using Linux networking tools.

System Environment

Operating System: Ubuntu Linux

Primary Interface: eth0

Tools Used: ip, ping, dhclient

Initial Network State Analysis

The network interface eth0 was found to be in UP state with an assigned IPv4 address. Initial connectivity tests using ping to Google DNS (8.8.8.8) were successful, confirming proper internet access.

Network Failure Simulation

The network interface eth0 was manually disabled using the ip link command. After disabling the interface, all network connectivity was lost and ping commands returned 'Network is unreachable', successfully simulating a network failure.

Interface Restoration Attempt

The interface was brought back to UP state. Although the IP address was present, network connectivity was not restored, indicating an issue beyond interface state.

Root Cause Analysis

The failure occurred due to the absence of a default gateway. Disabling the interface removed routing information, and DHCP did not automatically restore it.

Network Recovery

The issue was resolved by renewing the DHCP lease using dhclient, which restored the default route and internet connectivity.

Conclusion

This experiment demonstrated practical network troubleshooting skills, including interface inspection, failure simulation, routing analysis, and recovery techniques commonly used in DevOps environments.

Final Status

Network failure successfully simulated and resolved.